题目

在量子色动力学（QCD）的框架下，磁单极子解是一类重要的拓扑孤子结构。具体而言，这里所讨论的磁单极子是狄拉克（Dirac）磁单极子在非阿贝尔规范理论中的推广，即从阿贝尔U(1)规范群推广到非阿贝尔SU(2)规范群，从而扩展了经典磁单极子概念的应用范围。

我们考虑在四维闵可夫斯基（Minkowski）时空背景下，以SU(2)为规范群的杨-米尔斯（Yang-Mills）理论。该理论的拉格朗日量由以下表达式给出：

$$L = -\frac{1}{4}\int d^4x \sum_{a=1}^{3} F^a_{\mu\nu}F^{a,\mu\nu}$$

其中，闵可夫斯基时空的度规张量选取为：

$$\eta_{\mu\nu} = \eta^{\mu\nu} = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & -1 & 0 & 0 \\ 0 & 0 & -1 & 0 \\ 0 & 0 & 0 & -1 \end{pmatrix}$$

进一步，我们引入推广的Wu-Yang磁单极子*势，其具体形式为：

$$A^a_0 = 0, \quad A^a_i = \sum_{b=1}^{3} \epsilon_{iab}\frac{1}{g}\frac{x_b}{r^2}[1+\phi(r)]$$

这里，符号 ϵ_{iab} 表示三维空间中的全反对称张量（即Levi-Civita符号），$r^2 = (x^1)^2 + (x^2)^2 + (x^3)^2$ 为径向坐标的平方，而 g 为杨-米尔斯理论的耦合常数*。

基于上述设定，我们需要推导出函数 $\phi(r)$ 的二阶导数 $\phi(r)''$ 的具体表达式，该表达式应仅由 $\phi(r)$ 本身以及径向坐标 r 表示，假设 $\phi'(0) = 0$。

答案

$$\phi''(r) = \frac{\phi(r)\left[\phi^2(r) - 1\right]}{r^2}$$

解答

由Yang-Mills理论的拉氏量，可以推导出势能函数 $\phi(r)$ 满足的运动方程，这里假设 $\phi'(0) = 0$。在这一问题中，势没有电场，且不含时。首先写出规范势 A^a_i 的各个非零分量：
对于 $a = 1$：

$$A^1_i = \sum_{b=1}^{3} \epsilon_{i1b}\frac{1}{g}\frac{x_b}{r^2}[1+\phi(r)] \Rightarrow \begin{cases} A^1_2 = \epsilon_{213}\frac{1}{g}\frac{x_3}{r^2}[1+\phi(r)] = -\frac{1}{g}\frac{x_3}{r^2}[1+\phi(r)] \\ A^1_3 = \epsilon_{312}\frac{1}{g}\frac{x_2}{r^2}[1+\phi(r)] = +\frac{1}{g}\frac{x_2}{r^2}[1+\phi(r)] \\ A^1_0 = A^1_1 = 0 \end{cases}$$

对于 $a = 2$：

$$A^2_i = \sum_{b=1}^{3} \epsilon_{i2b}\frac{1}{g}\frac{x_b}{r^2}[1+\phi(r)] \Rightarrow \begin{cases} A^2_1 = \epsilon_{123}\frac{1}{g}\frac{x_3}{r^2}[1+\phi(r)] = +\frac{1}{g}\frac{x_3}{r^2}[1+\phi(r)] \\ A^2_3 = \epsilon_{321}\frac{1}{g}\frac{x_1}{r^2}[1+\phi(r)] = -\frac{1}{g}\frac{x_1}{r^2}[1+\phi(r)] \\ A^2_0 = A^2_2 = 0 \end{cases}$$

对于 $a = 3$：

$$A^3_i = \sum_{b=1}^{3} \epsilon_{i3b}\frac{1}{g}\frac{x_b}{r^2}[1+\phi(r)] \Rightarrow \begin{cases} A^3_1 = \epsilon_{132}\frac{1}{g}\frac{x_2}{r^2}[1+\phi(r)] = -\frac{1}{g}\frac{x_2}{r^2}[1+\phi(r)] \\ A^3_2 = \epsilon_{231}\frac{1}{g}\frac{x_1}{r^2}[1+\phi(r)] = +\frac{1}{g}\frac{x_1}{r^2}[1+\phi(r)] \\ A^3_0 = A^3_3 = 0 \end{cases}$$

考虑Yang-Mills 场强分量，场强张量的定义为：

$$F^a_{\mu\nu} = \partial_\mu A^a_\nu - \partial_\nu A^a_\mu - g\epsilon_{abc}A^b_\mu A^c_\nu$$

首先计算与 A^a_i 相关的导数 $\partial_i A^a_j$，即计算 $\partial_i\left[(1+\phi(r))/r^2\right]$。先考虑任意函数 $f(r)$ 的导数：

$$\partial_i f = \partial_r f\frac{\partial r}{\partial x_i} = \frac{1}{2\sqrt{x_1^2+x_2^2+x_3^2}}\cdot 2x_i\left(\partial_r f\right) = \frac{x_i}{r}\partial_r f$$

因此规范势中标量函数项的导数是：

$$
\begin{aligned}
\partial_i\left(\frac{1+\phi(r)}{r^2}\right) &= \frac{x_i}{r}\frac{\partial}{\partial r}\left(\frac{1+\phi(r)}{r^2}\right) = \frac{x_i}{r}\frac{r^2\phi'(r)-2r[1+\phi(r)]}{r^4} \\
&= \frac{x_i}{r^4}\left[r\phi'(r)-2\phi(r)-2\right]
\end{aligned}
$$

由规范势的表达式和 $F^a_{\mu\nu}$ 的定义式可知 $\mu=\nu=0$ 的项均为零。

色荷 $a=1$ 的分量，计算 F^1_{12}：

$$
\begin{aligned}
F^1_{12} &= \partial_1 A^1_2 - \partial_2 A^1_1 - g\epsilon_{1bc}A^b_1A^c_2 = \partial_1 A^1_2 - g\epsilon_{123}A^2_1A^3_2 - g\epsilon_{132}A^3_1A^2_2 \\
&= \partial_1 A^1_2 - gA^2_1A^3_2 \\
&= -\frac{x_3}{g}\partial_1\left(\frac{1+\phi(r)}{r^2}\right) - g\left[\frac{1}{g}\frac{x_3}{r^2}(1+\phi(r))\right]\left[\frac{1}{g}\frac{x_1}{r^2}(1+\phi(r))\right] \\
&= -\frac{x_1x_3}{gr^4}[r\phi'(r)-2\phi(r)-2] - \frac{x_1x_3}{gr^4}[1+\phi(r)]^2 \\
&= -\frac{x_1x_3}{gr^4}\left[r\phi'(r)+\phi^2(r)-1\right] = -F^1_{21}
\end{aligned}
$$

计算 F^1_{23}：

$$
\begin{aligned}
F^1_{23} &= \partial_2 A^1_3 - \partial_3 A^1_2 - g\epsilon_{1bc}A^b_2A^c_3 \\
&= \partial_2 A^1_3 - \partial_3 A^1_2 - g\epsilon_{123}A^2_2A^3_3 - g\epsilon_{132}A^3_2A^2_3 \\
&= \partial_2 A^1_3 - \partial_3 A^1_2 - gA^3_2A^2_3 \\
&= \frac{x_2^2}{g}\partial_r\left(\frac{1+\phi(r)}{r^2}\right) + \frac{x_3^2}{g}\partial_r\left(\frac{1+\phi(r)}{r^2}\right) + g\left[\frac{1}{g}\frac{x_1}{r^2}(1+\phi(r))\right]\left[-\frac{1}{g}\frac{x_1}{r^2}(1+\phi(r))\right] \\
&= \frac{r^2-x_1^2}{gr^4}\left(r\phi'(r)-2\phi(r)-2\right) - \frac{x_1^2}{gr^4}[1+\phi(r)]^2 + \frac{2r^2}{gr^4}[1+\phi(r)] \\
&= \frac{1}{gr^4}\left[\left(r^2-x_1^2\right)r\phi'(r) - 2\left(r^2-x_1^2\right)(\phi(r)+1) + 2r^2(\phi(r)+1) - x_1^2(\phi^2(r)+2\phi(r)+1)\right] \\
&= \frac{1}{gr^4}\left[\left(r^2-x_1^2\right)r\phi'(r) + 2x_1^2(\phi(r)+1) - x_1^2\phi^2(r) - x_1^2(2\phi(r)+1)\right] \\
&= \frac{1}{gr^4}\left[\left(r^2-x_1^2\right)r\phi'(r) - \left(\phi^2(r)-1\right)x_1^2\right] \\
&= -\frac{1}{gr^4}\left[r(x_1^2-r^2)\phi'(r) + \left(\phi^2(r)-1\right)x_1^2\right] = -F^1_{32}
\end{aligned}
$$

计算 F^1_{13}：

$$
\begin{aligned}
F^1_{13} &= \partial_1 A^1_3 - \partial_3 A^1_1 - g\epsilon_{1bc}A^b_1A^c_3 \\
&= \partial_1 A^1_3 - g\epsilon_{123}A^2_1A^3_3 - g\epsilon_{132}A^3_1A^2_3 \\
&= \partial_1 A^1_3 + gA^3_1A^2_3 \\
&= \frac{x_1x_2}{g}\partial_r\left(\frac{1+\phi(r)}{r^2}\right) + g\left[-\frac{1}{g}\frac{x_2}{r^2}(1+\phi(r))\right]\left[-\frac{1}{g}\frac{x_1}{r^2}(1+\phi(r))\right] \\
&= \frac{x_1x_2}{gr^4}[r\phi'(r)-2\phi(r)-2] + \frac{x_1x_2}{gr^4}[1+\phi(r)]^2 \\
&= \frac{x_1x_2}{gr^4}\left[r\phi'(r)+\phi^2(r)-1\right] = -F^1_{31}
\end{aligned}
$$

色荷 $a=2$ 的分量，计算 F^2_{12}：

$$
\begin{aligned}
F^2_{12} &= \partial_1 A^2_2 - \partial_2 A^2_1 - g\epsilon_{2bc}A^b_1A^c_2 = -\partial_2 A^2_1 - g\epsilon_{213}A^1_1A^3_2 - g\epsilon_{231}A^3_1A^1_2 \\
&= -\partial_2 A^2_1 - gA^3_1A^1_2 \\
&= -\frac{x_2x_3}{gr^4}[r\phi'(r)-2\phi(r)-2] - g\left[-\frac{1}{g}\frac{x_2}{r^2}(1+\phi(r))\right]\left[-\frac{1}{g}\frac{x_3}{r^2}(1+\phi(r))\right] \\
&= -\frac{x_2x_3}{gr^4}\left[r\phi'(r)+\phi^2(r)-1\right] = -F^2_{21}
\end{aligned}
$$

计算 F_{23}^2：

$$
\begin{aligned}
F_{23}^2 &= \partial_2 A_3^2 - \partial_3 A_2^2 - g\epsilon_{2bc} A_2^b A_3^c = \partial_2 A_3^2 - g\epsilon_{213} A_2^1 A_3^3 - g\epsilon_{231} A_2^3 A_3^1 \\
&= \partial_2 A_3^2 - g A_2^3 A_3^1 \\
&= -\frac{x_1 x_2}{gr^4}[r\phi'(r) - 2\phi(r) - 2] - g\left[\frac{1}{g}\frac{x_1}{r^2}(1+\phi(r))\right]\left[\frac{1}{g}\frac{x_2}{r^2}(1+\phi(r))\right] \\
&= -\frac{x_1 x_2}{gr^4}[r\phi'(r) + \phi^2(r) - 1] = -F_{32}^2
\end{aligned}
$$

计算 F_{13}^2：

$$
\begin{aligned}
F_{13}^2 &= \partial_1 A_3^2 - \partial_3 A_1^2 - g\epsilon_{2bc} A_1^b A_3^c \\
&= \partial_1 A_3^2 - \partial_3 A_1^2 - g\epsilon_{213} A_1^1 A_3^3 - g\epsilon_{231} A_1^3 A_3^1 \\
&= \partial_1 A_3^2 - \partial_3 A_1^2 - g A_1^3 A_3^1 \\
&= -\frac{x_1^2 + x_3^2}{gr^4}[r\phi'(r) - 2\phi(r) - 2] - g\left[-\frac{1}{g}\frac{x_2}{r^2}(1+\phi(r))\right]\left[\frac{1}{g}\frac{x_2}{r^2}(1+\phi(r))\right] + 2\frac{1+\phi(r)}{gr^2} \\
&= \frac{x_2^2 - r^2}{gr^4}[r\phi'(r) - 2\phi(r) - 2] + \frac{x_2^2}{gr^4}[1+\phi(r)]^2 + \frac{2r^2(1+\phi(r))}{gr^4} \\
&= \frac{1}{gr^4}\left[r\left(x_2^2 - r^2\right)\phi'(r) + \left(\phi^2(r) - 1\right)x_2^2\right] = -F_{31}^2
\end{aligned}
$$

色荷 $a = 3$ 的分量，计算 F_{12}^3：

$$
\begin{aligned}
F_{12}^3 &= \partial_1 A_2^3 - \partial_2 A_1^3 - g\epsilon_{3bc} A_1^b A_2^c \\
&= \partial_1 A_2^3 - \partial_2 A_1^3 - g\epsilon_{312} A_1^1 A_2^2 - g\epsilon_{321} A_1^2 A_2^1 \\
&= \partial_1 A_2^3 - \partial_2 A_1^3 - g A_1^2 A_2^1 \\
&= \frac{x_1^2 + x_2^2}{gr^4}[r\phi'(r) - 2\phi(r) - 2] + g\left[\frac{1}{g}\frac{x_3}{r^2}(1+\phi(r))\right]\left[-\frac{1}{g}\frac{x_3}{r^2}(1+\phi(r))\right] + 2\frac{1+\phi(r)}{gr^2} \\
&= -\frac{x_3^2 - r^2}{gr^4}[r\phi'(r) - 2\phi(r) - 2] - \frac{x_3^2}{gr^4}[1+\phi(r)]^2 + \frac{2r^2(1+\phi(r))}{gr^4} \\
&= -\frac{1}{gr^4}[r(x_3^2 - r^2)\phi'(r) + (\phi^2(r) - 1)x_3^2] = -F_{21}^3
\end{aligned}
$$

计算 F_{23}^3：

$$
\begin{aligned}
F_{23}^3 &= \partial_2 A_3^3 - \partial_3 A_2^3 - g\epsilon_{3bc} A_2^b A_3^c \\
&= -\partial_3 A_2^3 - g\epsilon_{312} A_2^1 A_3^2 - g\epsilon_{321} A_2^2 A_3^1 \\
&= -\partial_3 A_2^3 - g A_2^1 A_3^2 \\
&= -\frac{x_1 x_3}{gr^4}[r\phi'(r) - 2\phi(r) - 2] - g\left[-\frac{1}{g}\frac{x_3}{r^2}(1+\phi(r))\right]\left[-\frac{1}{g}\frac{x_1}{r^2}(1+\phi(r))\right] \\
&= -\frac{x_1 x_3}{gr^4}[r\phi'(r) - 2\phi(r) - 2] - \frac{x_1 x_3}{gr^4}[1+\phi(r)]^2 \\
&= -\frac{x_1 x_3}{gr^4}[r\phi'(r) + \phi^2(r) - 1] = -F_{32}^3
\end{aligned}
$$

计算 F_{13}^3：

$$
\begin{aligned}
F_{13}^3 &= \partial_1 A_3^3 - \partial_3 A_1^3 - g\epsilon_{3bc} A_1^b A_3^c \\
&= -\partial_3 A_1^3 - g\epsilon_{312} A_1^1 A_3^2 - g\epsilon_{321} A_1^2 A_3^1 \\
&= -\partial_3 A_1^3 + g A_1^2 A_3^1 \\
&= \frac{x_2 x_3}{gr^4}[r\phi'(r) - 2\phi(r) - 2] + g\left[\frac{1}{g}\frac{x_3}{r^2}(1+\phi(r))\right]\left[\frac{1}{g}\frac{x_2}{r^2}(1+\phi(r))\right] \\
&= \frac{x_2 x_3}{gr^4}[r\phi'(r) - 2\phi(r) - 2] + \frac{x_2 x_3}{gr^4}[1+\phi(r)]^2 \\
&= \frac{x_2 x_3}{gr^4}[r\phi'(r) + \phi^2(r) - 1] = -F_{31}^3
\end{aligned}
$$

由此得到全部非零的场强分量。由指标升降的法则：

$$F^{a,\mu\nu} = \eta^{\mu\rho}\eta^{\nu\sigma}F^a_{\rho\sigma}$$

这里非零的 $F^a_{\mu\nu}$ 中，$\mu,\nu = 1,2,3$，因此：

$$F^{a,ij} = \eta^{ik}\eta^{jl}F^a_{kl} = (-\delta^{ik})(-\delta^{jl})F^a_{kl} = F^a_{ij}$$

因此，拉氏密度的表达式展开写为：

$$\mathcal{L}_{\mathrm{YM}} = \sum_{a=1}^{3} F^a_{\mu\nu}F^{a,\mu\nu}$$

$$= 2\left[(F^1_{12})^2 + (F^1_{13})^2 + (F^1_{23})^2\right] + 2\left[(F^2_{12})^2 + (F^2_{13})^2 + (F^2_{23})^2\right] + 2\left[(F^3_{12})^2 + (F^3_{13})^2 + (F^3_{23})^2\right]$$

$$\equiv \mathcal{L}_1 + \mathcal{L}_2 + \mathcal{L}_3$$

代表三种色荷部分的求和。

以第一项 $a = 1$ 为例：

$$\begin{aligned}
\mathcal{L}_1 &= 2\left[(F^1_{12})^2 + (F^1_{13})^2 + (F^1_{23})^2\right] \\
&= \frac{2}{g^2r^8}\left\{\left[r\phi'(r) + \phi^2(r) - 1\right]^2\left(x_1^2x_3^2 + x_1^2x_2^2\right) + \left[r(x_1^2 - r^2)\phi'(r) + \left(\phi^2(r) - 1\right)x_1^2\right]^2\right\} \\
&= \frac{2}{g^2r^8}\left[\left(r^2\phi'^2(r) + \left(\phi^2(r) - 1\right)^2 + 2r\phi'(r)(\phi^2(r) - 1)\right)x_1^2\left(r^2 - x_1^2\right) + \left(r^2\phi'^2(r)(x_1^2 - r^2)^2 + x_1^4(\phi^2(r) - 1)^2 + 2r\left(x_1^2 - r^2\right)\phi'(r)x_1^2\right.\right. \\
&= \frac{2}{g^2r^8}\left[r^2\phi'^2(r)x_1^2\left(r^2 - x_1^2\right) + x_1^2(\phi^2(r) - 1)^2\left(r^2 - x_1^2\right) + r^2\left(x_1^2 - r^2\right)^2\phi'^2(r) + x_1^4(\phi^2(r) - 1)^2 + 2r\left(x_1^2 - r^2\right)\phi'(r)x_1^2\left(\phi^2(r) - 1\right)\right] \\
&= \frac{2}{g^2r^8}\left\{r^2\phi'^2(r)\left(r^2 - x_1^2\right)\left[x_1^2 + \left(r^2 - x_1^2\right)\right] + x_1^2(\phi^2(r) - 1)^2\left[x_1^2 + \left(r^2 - x_1^2\right)\right]\right\} \\
&= \frac{2}{g^2r^8}\left[r^4\phi'^2(r)\left(r^2 - x_1^2\right) + r^2x_1^2\left(\phi^2(r) - 1\right)^2\right] \\
&= \frac{2}{g^2r^6}\left[r^2\phi'^2(r)\left(r^2 - x_1^2\right) + x_1^2(\phi^2(r) - 1)^2\right]
\end{aligned}$$

第二、三项 $a = 2,3$ 的结果很显然与 $a = 1$ 非常相似：

$$\begin{aligned}
\mathcal{L}_2 &= \frac{2}{g^2r^6}\left[r^2\phi'^2(r)\left(r^2 - x_2^2\right) + x_2^2\left(\phi^2(r) - 1\right)^2\right] \\
\mathcal{L}_3 &= \frac{2}{g^2r^6}\left[r^2\phi'^2(r)\left(r^2 - x_3^2\right) + x_3^2\left(\phi^2(r) - 1\right)^2\right]
\end{aligned}$$

拉氏密度的最终结果：

$$\begin{aligned}
\mathcal{L}_{\mathrm{YM}} &= \frac{2}{g^2r^6}\left\{r^2\phi'^2(r)\left[\left(r^2 - x_1^2\right) + \left(r^2 - x_2^2\right) + \left(r^2 - x_3^2\right)\right] + \left(x_1^2 + x_2^2 + x_3^2\right)\left(\phi^2(r) - 1\right)^2\right\} \\
&= \frac{2}{g^2r^6}\left[2r^4\phi'^2(r) + r^2\left(\phi^2(r) - 1\right)^2\right] = \frac{2}{g^2r^4}\left[2r^2\phi'^2(r) + \left(\phi^2(r) - 1\right)^2\right] \\
&= \frac{4}{g^2r^2}\phi'^2(r) + \frac{2}{g^2r^4}\left(\phi^2(r) - 1\right)^2
\end{aligned}$$

将拉氏量中的四维时空积分在球坐标系下改写，并且由于被积函数所具有的球对称性，与角度有关的积分可以直接算出：

$$\int d^4x = \int dt\, d^3x = \int dt \int_0^{2\pi} d\varphi \int_0^{\pi} \sin\theta\, d\theta \int_0^{\infty} r^2 dr = 4\pi \int dt \int_0^{\infty} dr\, r^2$$

拉氏量的结果是：

$$
\begin{aligned}
L &= -\frac{1}{4} \times 4\pi \int dt \int_0^\infty r^2 \left[\frac{4}{g^2 r^2} \phi'^2(r) + \frac{2}{g^2 r^4} \left(\phi^2(r) - 1 \right)^2 \right] dr \\
&= -\frac{4\pi}{g^2} \int dt \int_0^\infty dr \left[\frac{\left(\phi^2(r) - 1 \right)^2}{2r^2} + \phi'^2(r) \right]
\end{aligned}
$$

对 $\phi(r)$ 的变分将给出相应的 Euler-Lagrange 方程：

$$
\frac{\partial \mathcal{L}}{\partial \phi(r)} - \frac{d}{dr} \frac{\partial \mathcal{L}}{\partial \phi'(r)} = 0
$$

其中导数分别是：

$$
\frac{\partial \mathcal{L}}{\partial \phi(r)} = \frac{2 \left[\phi^2(r) - 1 \right] \phi(r)}{r^2}, \quad \frac{\partial \mathcal{L}}{\partial \phi'(r)} = 2\phi'(r)
$$

综上所述，势能函数 $\phi(r)$ 满足的运动方程是：

$$
\phi''(r) = \frac{\phi(r) \left[\phi^2(r) - 1 \right]}{r^2}
$$